

SERTAR, China

WMO: 561520

Lat: 32.2749N

Lon: 100.3337E

Elev: 12782

StdP: 9.06

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.6	(c) -4.1	(d) -20.4	(e) 2.9	(f) 6.7	(g) -17.2	(h) 3.5	(i) 8.7	(j) 26.4	(k) 31.6	(l) 23.2	(m) 32.3	(n) 1.8	(o) 290	(p) 0.724

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.0	(c) 67.5	(d) 51.2	(e) 65.2	(f) 50.2	(g) 62.9	(h) 49.3	(i) 53.5	(j) 63.8	(k) 52.2	(l) 62.1	(m) 51.2	(n) 60.3	(o) 5.7	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 49.6	(b) 85.8	(c) 58.0	(d) 48.5	(e) 82.5	(f) 56.4	(g) 47.6	(h) 79.5	(i) 55.3	(j) 28.2	(k) 64.0	(l) 27.2	(m) 62.5	(n) 26.4	(o) 60.6	(p) 61.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 20.9	(b) 17.3	(c) 14.3	(e) -14.4	(f) 72.1	(g) 3.6	(h) 1.5	(i) -17.0	(j) 73.1	(k) -19.0	(l) 74.0	(m) -21.1	(n) 74.9	(o) -23.6	(p) 76.0		
(a) 20.9	(b) 17.3	(c) 14.3	(e) -15.2	(f) 56.0	(g) 3.3	(h) 1.8	(i) -17.6	(j) 57.3	(k) -19.5	(l) 58.3	(m) -21.3	(n) 59.3	(o) -23.7	(p) 60.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.5	16.2	21.0	27.6	36.1	43.2	49.4	52.2	51.3	47.1	37.6	25.8	18.0		
	DBStd	13.73	5.47	5.46	5.29	4.46	4.71	4.47	3.71	3.69	4.68	5.66	5.76	5.28		
	HDD50	5514	1049	812	693	417	216	64	22	30	108	384	727	992		
	HDD65	10754	1514	1232	1158	867	674	467	397	426	537	848	1177	1457		
	CDD50	236	0	0	0	0	7	47	91	69	21	1	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	4.7	5.2	5.5	5.5	5.3	5.2	4.6	4.3	3.8	3.9	4.0	4.2	4.5		
	PrecAvg	25.6	0.2	0.3	0.5	0.9	2.4	5.2	5.7	4.1	4.6	1.3	0.2	0.2		
(15) Precipitation	PrecMax	31.6	0.6	0.7	1.5	2.0	4.1	8.6	10.0	6.9	7.1	3.2	0.8	0.5		
	PrecMin	20.3	0.0	0.0	0.0	0.2	0.5	2.3	2.2	1.3	1.6	0.2	0.0	0.0		
	PrecStd	2.6	0.2	0.2	0.4	0.4	0.8	1.4	1.9	1.4	1.4	0.7	0.2	0.2		
	PrecStd	2.6	0.2	0.2	0.4	0.4	0.8	1.4	1.9	1.4	1.4	0.7	0.2	0.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.8	48.9	52.9	60.0	66.5	69.0	70.2	70.1	67.4	63.0	51.2	46.2		
		MCWB	29.0	30.0	33.9	38.3	45.7	51.3	53.4	52.2	50.0	44.7	32.3	29.1		
	2%	DB	40.3	44.0	48.2	55.3	62.3	64.8	67.3	67.1	64.0	57.0	46.9	41.5		
		MCWB	25.6	27.7	31.4	36.7	43.7	50.0	52.1	51.4	49.2	42.5	30.6	26.6		
	5%	DB	36.1	40.3	44.6	51.9	58.7	62.0	64.7	64.3	60.7	52.6	43.5	37.4		
		MCWB	23.3	26.2	29.8	35.1	42.1	49.2	51.3	50.5	48.3	39.7	29.2	24.6		
	10%	DB	32.1	36.6	41.3	48.5	55.3	59.3	62.0	61.6	57.5	48.8	40.0	33.6		
		MCWB	21.3	24.4	28.4	33.9	41.0	47.8	50.4	49.6	46.7	37.8	27.6	22.6		
	0.4%	WB	29.5	30.4	35.1	40.9	48.6	53.9	55.7	54.6	52.5	46.8	33.5	29.9		
		MCDB	45.1	46.6	47.8	53.2	61.7	64.3	66.6	65.6	63.5	59.4	46.1	44.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	26.0	28.3	32.5	38.6	46.2	52.0	53.9	53.0	50.6	43.9	31.6	27.0		
		MCDB	39.4	42.5	44.9	50.9	58.6	62.2	63.9	63.7	60.9	54.4	44.3	39.9		
	5%	WB	23.6	26.5	30.8	36.8	44.3	50.5	52.6	51.8	49.2	41.4	30.0	25.0		
		MCDB	35.1	39.1	42.0	48.0	55.3	60.0	62.1	61.9	58.7	49.6	41.4	36.5		
	10%	WB	21.6	24.8	29.2	35.3	42.4	49.0	51.4	50.6	47.7	39.2	28.4	22.9		
		MCDB	31.5	35.0	39.3	45.7	52.0	57.4	60.0	59.6	56.2	46.3	38.6	32.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	33.2	30.7	26.1	22.2	21.5	18.3	19.0	20.3	20.7	21.6	28.1	32.8		
		MCDBR	39.6	37.9	32.5	30.3	28.6	22.7	24.9	26.9	25.7	27.4	33.2	38.3		
		MCWBR	26.3	23.8	17.6	13.8	12.2	10.1	11.5	12.6	12.4	13.3	18.4	24.9		
		MCDBR	37.1	34.6	27.7	24.0	22.8	19.5	19.6	21.6	22.4	23.0	29.6	36.5		
	5% WB	MCWBR	25.1	22.0	15.3	11.7	10.6	9.4	9.5	10.5	11.2	11.7	16.9	24.0		
		MCWBR	25.1	22.0	15.3	11.7	10.6	9.4	9.5	10.5	11.2	11.7	16.9	24.0		
(40) Clear-Sky Solar Irradiance	taub		0.212	0.240	0.290	0.309	0.300	0.288	0.289	0.288	0.264	0.239	0.213	0.198		
	taud		2.588	2.510	2.348	2.341	2.451	2.596	2.613	2.597	2.670	2.668	2.652	2.657		
	Ebn at Noon		332	328	315	311	312	314	313	314	321	325	330	334		
	Edh at Noon		24	30	38	40	36	31	30	30	27	25	23	22		
	RadAvg		1215	1483	1820	2065	1923	1718	1780	1690	1499	1331	1296	1171		
(45) All-Sky Solar Radiation	RadStd		89	73	92	169	147	147	120	182	114	72	67	63		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.94	+1.93	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	+0.72	+1.01	-1.46	+1.13	N/A	N/A	-187	-302	+110	N/A

Nomenclature: See separate page